

CHAPTER 13

Qualitative Studies

A qualitative study looks for the meanings, themes, and theories that emerge from observing and interacting with study participants.

13.1 Overview

Quantitative research uses structured, hypothesis-driven approaches to gather data that can be statistically analyzed. **Qualitative research** uses in-depth interviews, focus group discussions, participant observation,

and other unstructured or semi-structured methods to explore attitudes and perceptions, identify themes and patterns, and formulate new theories. Qualitative research seeks to answer questions like “why?” and “how?” that numbers-focused quantitative research cannot adequately answer (**Figure 13-1**).

FIGURE 13-1 Comparing Typical Qualitative and Quantitative Research Approaches

	Quantitative	Qualitative
Main types of questions answered	Who? Where? When? What?	Why? How?
Participants	A large randomly sampled population	A small purposefully recruited population
Data collection approach	Structured	Unstructured or semi-structured
Common data collection methods	Questionnaires	In-depth interviews, focus groups, observation
Types of questions asked	Closed-ended (fixed response options)	Open-ended (flexible response options)
Data collected	Numeric data	Textual data (words, images, and objects)
Goal of data analysis	Test existing hypotheses	Formulate new theories
Outcomes reported	Statistics	Themes and theories

The word **phenomenon** is often used to describe the central concept being studied during a qualitative research project. In the health sciences, many qualitative research projects aim to improve health promotion programs and clinical processes or to provide a foundation for social change. Other qualitative studies seek to understand how people experience health and illness as individuals and as members of communities, why they engage in or do not engage in various health-related behaviors, and how they make health-related decisions.

13.2 Ontology, Epistemology, and Axiology

Because qualitative research examines thoughts and beliefs that cannot be directly measured, researchers using qualitative methods must be aware of the ways their own thoughts, experiences, biases, and assumptions shape the design and interpretation of their research studies. The philosophical orientations that guide both quantitative and qualitative research are rooted in ontology, epistemology, and axiology (Figure 13-2).

Ontology is the study of the nature of reality and truth. Ontological perspectives extend from realism to relativism. **Realism** assumes

that one reality exists, and it can be understood. **Objectivity** describes facts that can be evaluated without bias. A realist would say that meaning exists in an object independent of the subject investigating it. Quantitative research studies seek to make objective determinations about the world. **Relativism** assumes that there are multiple realities and they cannot be fully understood. **Subjectivity** describes claims and experiences that are interpreted based on an evaluator's beliefs, perceptions, and feelings. A relativist would say that a subject imposes meaning on an object. Qualitative research designs typically seek to understand subjective aspects of human existence.

Epistemology is the study of knowledge. Ontology explains how a researcher defines reality and truth, while epistemology explains how a researcher knows what is real and true. For example, a guiding principle for quantitative studies is **empiricism**, the assumption that the senses (such as seeing, hearing, and touching) are the best way to measure truth about the world. Quantitative research assumes that researchers are independent from their study subjects and that researchers can control for possible biases in order to make objective measurements. Qualitative research assumes that researchers and study participants are interdependent and create knowledge together as they interactively explore subjective topics.

FIGURE 13-2 Ontology, Epistemology, and Axiology

		Quantitative Studies	Qualitative Studies
Ontology	What is reality?	Realism (one reality exists)	Relativism (multiple realities exist)
		Objectivism	Subjectivism
Epistemology	What is knowledge?	Positivism (reality can be measured)	Constructivism/interpretivism (reality must be created and interpreted)
Axiology	What are values?	Inquiry should be value-free, context-free, and unbiased	Inquiry is value-laden, context-bound, and biased

Axiology, or value theory, is the study of values. Quantitative researchers usually apply an axiological perspective that assumes that rigorous procedures can eliminate the impact of values and biases on study outcomes. Qualitative researchers assume that a researcher's values affect the study. Researchers using qualitative approaches typically take time to examine their own backgrounds, beliefs, perspectives, and biases so that they can make informed decisions about the methods they use, their own role in the study, the ways in which they interact with study participants, and how they interpret data.

13.3 Theoretical Paradigms

Because the goal of qualitative research is to understand a complex phenomenon rather than to predict some observable event, qualitative researchers must select the theoretical perspectives that will guide their study design, data analysis, and interpretation. Four of the most prominent philosophical assumptions applied to health science research are

positivism, constructivism/interpretivism, critical theory, and pragmatism (Figure 13-3).

Positivism is a paradigm in which researchers apply a realist perspective that assumes that reality is knowable and that inquiry should be logical and value-free. The word "positive" denotes definitiveness and certainty. Studies based on a positivist paradigm generally use empirical methods to test hypotheses and predict outcomes. They often employ quantitative experimental designs. **Post-positivism** is a related paradigm in which researchers aim to experimentally test theories about how the world works, but they acknowledge that the unpredictability of human behavior limits the validity of some empirical methods. Post-positivism applies a critical realist perspective to research. Studies using a post-positivist paradigm often employ quasi-experimental and mixed methods designs.

Constructivism is a paradigm in which researchers have a relativist perspective that considers each individual's reality to be a function of that person's lived experiences. The ontological assumption of constructivism is that there are many realities, not just one

FIGURE 13-3 Examples of Theoretical Paradigms

	Positivism	Constructivism/ Interpretivism	Critical Theory	Pragmatism
Ontology	There is a single reality (realism).	Reality is constructed by individuals in groups (relativism).	Realities are socially and historically constructed.	There is a single reality, but individuals interpret reality differently.
Epistemology	Reality can be measured (objectivism).	Reality needs to be interpreted (subjectivism).	Reality and knowledge are influenced by power relationships within societies.	The best method for understanding reality is the one that solves problems.
Common methodologies	Experimental research and other quantitative methods	Qualitative methods	Qualitative methods	Mixed methods

reality, and realities are created as researchers and participants interact. The epistemological assumption is that researchers and participants must work together to understand reality. The axiological assumption is that beliefs and values are social constructs. These philosophical stances make constructivism the opposite of positivism. **Interpretivism** is a related paradigm in which researchers consider the reality in the social world to be different from reality in the natural world. Interpretivism and constructivism are about understanding how various groups of people interpret reality. Studies based on constructivism and interpretivism use qualitative study designs.

Critical theory is a paradigm that considers reality to be dependent on social and historical constructs and assumes that reality can be uncovered by identifying and challenging power structures. The philosophical assumption of critical theory is that participants can actively construct realities that are shaped by beliefs and values. Research conducted under a critical theory framework is change-oriented. Studies based on critical theory often use participatory methods that empower participants and equip them to advocate for change. For example, studies based on critical theory may use an **action research** approach in which participants work together to solve a social problem. Action research conducted under a **transformative paradigm** assumes that reality can be changed when researchers collaborate with participants from marginalized populations to address a social justice issue.

Pragmatism is a paradigm in which researchers assume that reality is situational, and it is acceptable to use any and all research tools and frameworks to try to understand a particular problem so it can be solved. The goal of pragmatic research is to solve problems, so the focus is on the outcomes of the research project rather than the theories and processes that guide it. The ontological assumption of pragmatism is **symbolic realism**, which treats individuals' realities as being real to those individuals, because that

orientation is the most useful and practical. The epistemological assumption is that reality can be known and understood using many different approaches. The axiological assumption is that beliefs and values are part of practical decisions. Pragmatists often use mixed methods in their study designs.

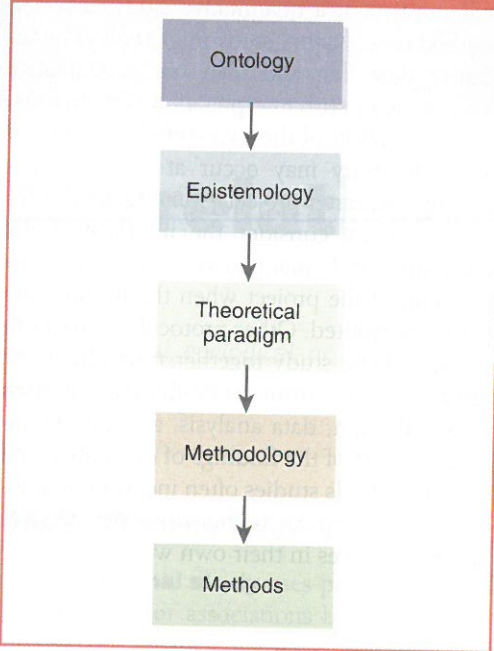
Theoretical **pluralism** occurs when a researcher draws on more than one theoretical framework to guide the design, analysis, and interpretation of a research project. Researchers using two or more theoretical perspectives must carefully explain why the theories were selected, how they were implemented, and how the use of multiple theories informed the insights gained from the study.

13.4 Qualitative Methodologies

The methodologic approaches selected for a qualitative research project must align with the goals of the study and the selected theoretical paradigm. Planning for a qualitative study often progresses from the identification of the underlying ontology and epistemology to the selection of a suitable paradigm and then the identification of methodology that aligns with the underlying theoretical perspective (Figure 13-4). In the health sciences, the most popular qualitative methodologies include phenomenology, grounded theory, ethnography, and case studies (Figure 13-5).

Phenomenology seeks to understand how individuals interpret and find meaning in their own unique life experiences and feelings. The researcher uses in-depth interviews to gather data from several people, then the transcripts of those interviews are examined so that meanings and themes can be identified and understood from the perspective of the participants. **Bracketing** is the process of a researcher intentionally setting aside any preconceived ideas about reality in order to be open to new meanings that might be expressed by participants.

FIGURE 13-4 Qualitative Research Planning Process



Grounded theory is an inductive reasoning process that uses observations to develop general theories that explain human behavior or other phenomena. Data collection and data analysis occur simultaneously, so that theories can be developed and refined. **Theoretical sampling** uses the emerging theory to guide the selection of new data sources. Data collection continues until data saturation has been reached.

FIGURE 13-5 Examples of Qualitative Methodologies

	Phenomenology	Grounded Theory	Ethnography	Case Studies
Goal	Understand how individuals gain meaning from lived experiences	Construct a theoretical model to explain a phenomenon	Understand the collective experience of sociocultural groups	Understand an event, process, or program
Common methods	In-depth interviews	Interviews and observations	Participant observation	Observations, interviews, documents, and other reports

Ethnography is the systematic study of people and cultures in their natural environments. Ethnography is an anthropological approach in which researchers aim to develop an insider's view (an **emic perspective**), rather than an outsider's view (an **etic perspective**), of how members of a particular sociocultural group understand their world. Ethnographers often use participant observation methods to understand a group's collective experiences, values, beliefs, and behaviors. It is typical for ethnographers to immerse themselves in the study community and to intentionally interact with the group for many months or years.

A **case study** is a qualitative research approach that uses multiple data sources to examine one person, group, event, or other situation in detail. A case study approach may be used when preparing a case report or a case series, or it may be applied to understanding an event, process, or program. The goal is not to develop generalizable knowledge, but to understand one event well. Multiple data sources may be used as part of a holistic examination of the case, including observations, interviews, and reviews of historic documents.

13.5 Mixed Methods Research

Qualitative research can be complete on its own, but sometimes the best option is to use qualitative methods in conjunction with

quantitative surveys. **Mixed methods** projects use both quantitative and qualitative methods in one research study. Some mixed methods projects use a convergent parallel design to collect quantitative and qualitative data concurrently and then compare the results and interpret them. Some studies collect the data sequentially, completing one type of study first and then designing and implementing the other type of study. Some qualitative studies are embedded within a quantitative study.

Social science research can often be classified as being exploratory, descriptive, or explanatory. **Exploratory research** aims to discover new ideas and develop hypotheses. Descriptive research seeks to understand key characteristics of a group. **Explanatory research** tests hypotheses about causal relationships. A mixed methods exploratory study

might collect qualitative data first and then use the insights from that study to design and implement a quantitative study. A mixed methods explanatory study might collect quantitative data first and then use a qualitative study to assist with interpretation of the results.

Integration of the two strands of a mixed methods study may occur at various times during a research project. Some research protocols do not consider the interface of the qualitative and quantitative results until the very end of the project when the findings are being interpreted. Other protocols weave both strands of the study together throughout the research process from the design stage through data collection, data analysis, and interpretation. Reports of the findings of qualitative and mixed methods studies often incorporate quotations that express participants' perspectives and experiences in their own words.